

RENEWABLE ENERGY INTELLIGENCE

SolarShare

Peer-to-peer solar energy trading for neighbourhoods

SolarShare helps a rooftop solar owner offer expected surplus to nearby households. The system suggests how much can be shared, matches local demand, adjusts the price when supply or grid conditions change, and settles the result against final meter readings.

THEME	PROBLEM STATEMENT	PROTOTYPE
Renewable Energy Intelligence	Renewable Energy P2P Trading Marketplace	Responsive web application

1. Problem statement and proposed solution

Problem statement

Households with rooftop solar often overproduce during peak sunlight hours and are forced to export surplus energy back to the grid at low feed-in tariffs, while nearby consumers may still be paying high retail rates. The challenge asks for a peer-to-peer energy trading platform where prosumers can sell surplus solar directly to nearby consumers, with dynamic pricing based on real-time supply, demand, and grid congestion.

The gap

A solar household can receive a low export tariff for extra generation at the same time that a nearby household buys electricity at a higher retail rate. SolarShare creates a local market between those two prices. The seller can receive a better return, while the buyer can pay less than the retail rate.

Proposed solution

SolarShare is a neighbourhood marketplace for local solar energy credits. It estimates a home's surplus, asks the owner to approve a suggested amount, and lets nearby households reserve it. The price responds to supply, demand, and a neighbourhood congestion signal. After the sharing window, meter readings are used to settle the final amount.

For the prototype, solar production is modeled from public data and household meter readings are simulated. This is shown clearly in the interface. A deployed system would replace the simulator with authorized smart meter or inverter data.

Users

USER	NEED
Rooftop solar owner	A simple way to share surplus without preparing technical listings.
Nearby consumer	Access to local solar at a price below the configured retail tariff.
Utility or local operator	Visibility into local supply, demand, feeder conditions, and settlement.
Regulator	Clear pricing rules, consent records, and an auditable transaction history.

2. How SolarShare works

Seller experience

The default experience is Guided Sharing. The seller sees a plain message such as, "You may have 4 kWh of extra solar tomorrow between 1 PM and 3 PM." The estimate protects expected home use and any reserve set by the owner. The seller can approve it, decline it, or edit the amount. Manual listing stays available, and Auto Share is optional.

Buyer experience

"Solar is available nearby. Reserve up to 3 kWh and save about INR 6." The buyer chooses an amount and sees the expected price before confirming. Settlement uses the final available energy. The prototype uses energy credits and does not process real payments.

Transaction flow

- 1 Public weather data and a PV model estimate solar generation for the selected location and panel setup.
- 2 The system subtracts expected home demand and the owner's reserve, then suggests a shareable amount.
- 3 Nearby consumers place reservations. The matcher allocates the available solar within the neighbourhood.
- 4 The price changes with supply, demand, and the feeder congestion scenario, but stays within the configured feed-in and retail tariffs.
- 5 Final simulated meter readings reconcile the reservation. The ledger records seller credit, buyer cost, and the settled energy.

Pricing behaviour

MARKET CONDITION	PRICE AND MATCHING RESPONSE
High solar supply and low demand	Price moves closer to the feed-in tariff.
Limited solar and high demand	Price rises but remains below the retail tariff.
High feeder congestion	The matcher gives priority to local trades that reduce grid import during that period.

SolarShare records an energy and financial settlement. It does not claim that a particular unit of electricity is physically routed from one house to another.

3. Data and technical approach

The prototype uses public data where it is useful and simulation where household access would otherwise depend on a private device account. Each screen identifies the type of data being shown.

DATA	PROTOTYPE SOURCE	STATUS
Weather and solar conditions	Open-Meteo	Public data
PV generation	PVWatts v8 or PVGIS	Modeled
Home demand and final meter readings	Scenario generator	Simulated
Feeder capacity and load	Neighbourhood scenario	Simulated
Indian grid context	Published CEA data	Published data
Meter and inverter telemetry	Authorized device connector	Planned integration

System design

LAYER	RESPONSIBILITY
Web application	Seller, buyer, and operator views in one responsive interface.
Application API	Forecasts, listings, reservations, prices, settlement, and impact data.
Forecast service	Combines weather, panel settings, and expected household demand.
Market service	Calculates the price, matches reservations, and responds to congestion.
Settlement ledger	Keeps an append-only record of market events and final energy credits.
Data adapters	Connect public APIs now and approved meter or inverter services later.

Implementation choice

The first version uses a standard web client, REST APIs, a relational database, and a small rules-based market engine. Blockchain is listed in the problem statement as an optional technology. It is not needed to prove the main workflow, so the prototype uses an auditable database ledger instead.

Path to live household data

A production connector would ask for the household's consent, authenticate with an approved meter or inverter service, normalize the readings, and reject missing or implausible values before settlement. Vendor adapters or a standard such as SunSpec Modbus can sit behind this connector.